PRASHAST AWASTHI

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EDUCATION

Vellore Institute of Technology Vellore, India

B.Tech in Computer Science and Engineering, CGPA: 8.09 / 10.0 *Graduating: May 2027*

Relevant Coursework: Big Data Analytics, Probability and Statistics, Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, JavaScript
Data Science & Analytics: Pandas, NumPy, PostgreSQL, EDA, time-series forecasting, feature engineering, walk-forward cross-validation, Apache Spark, Plotly, Matplotlib
Machine Learning: Scikit-learn, XGBoost, LightGBM, Random Forest, SHAP, hyperparameter tuning, ablation testing
AI, Deep Learning & GenAI: HuggingFace Transformers, FinBERT, sentence-transformers, OpenCV, MediaPipe, LLM integration, RAG and agentic RAG, hybrid search, reranking, prompt engineering, multi-agent orchestration, tool use, function calling, guardrails, LangGraph, CrewAI, LlamaIndex, ChromaDB, FAISS, pgvector, Groq, Gemini, Ollama
Tools: FastAPI, REST APIs, Docker, Git/GitHub, Linux, AWS (EC2, S3, RDS), Streamlit, React, Pydantic, pytest

EXPERIENCE

Neve Jewels Mumbai, India

Data Science Intern *May 2026 – July 2026*

- Owned a next-day order forecasting problem end to end for an omnichannel jewellery business with 3 brands, selecting XGBoost (18.7% MAPE) over Linear Regression, Random Forest, Prophet, SARIMA, and LightGBM on 2.5 years of daily data, validated with a walk-forward cross-validation harness.
- Traced a forecast accuracy problem to a pipeline defect: a left join with campaign data followed by null removal silently dropped non-promotional days from training, biasing the model toward promotions; fixed the logic and restored full coverage.
- Engineered 25+ features across lag, rolling-statistic, fiscal calendar, holiday, discount, and marketing groups, retaining each group only when ablation showed a validation gain and it had a clear business rationale.
- Wrote SQL and Python preparation workflows joining order, discount, marketing, and campaign data; interpreted the model with SHAP (7-day lagged orders and discount depth strongest) and delivered a stakeholder report with Plotly visualizations.

AceQuant Research Remote

Quantitative Development Intern *May 2025 – July 2025*

- Designed and implemented 4+ systematic trading strategies (SMA, EMA, Bollinger Bands, MACD) in Python, encoding entry and exit rules, signal logic, and position thresholds.
- Backtested each strategy on historical data from 2 broker APIs and tuned thresholds systematically, comparing variants across 6+ performance metrics to identify consistently profitable configurations.

PROJECTS

Multi-Agent Financial Research Assistant | *Python, LangGraph, FinBERT, ChromaDB, Groq, FastAPI*

- Built an 8-agent equity research system with role specialization, in which an orchestrator delegates analyst queries to specialist agents (technical, fundamental, risk, sentiment) and merges their outputs into one research report.
- Designed a verification agent acting as a numeric guardrail: it recomputes every quantitative claim from source values and rejects out-of-tolerance results through a bounded self-correction loop.
- Classified sentiment across 5,000+ financial news articles with FinBERT, evaluating sentiment-score correlations against price behaviour across 10+ equities; built 22 modules with a passing pytest suite and a local Ollama fallback behind a common provider interface.

Smart Executive Briefing Assistant | *Python, LangGraph, PostgreSQL + pgvector, sentence-transformers, FastAPI*

- Developed an agentic AI tool that generates structured, source-cited executive briefings from 5 inputs, designed to replace a 30–45 min manual research workflow for consulting teams.
- Architected a 6-node LangGraph pipeline with a hybrid RAG retrieval stack (vector search, full-text search, reranking) over a 500+ document knowledge base combining SEC EDGAR filings, GDELT news, and web search.
- Prevented hallucinated figures by having the LLM cite metric reference IDs instead of writing numerals, running all numeric checks in deterministic Python and deleting unsupported claims; routed mechanical nodes to a fast model and reserved a frontier model for synthesis to cut token cost.

Face Recognition Photo Retrieval System | *Python, InsightFace, FAISS, FastAPI, React*

- Built a facial-embedding retrieval pipeline over a multi-thousand-image event archive, indexing InsightFace embeddings in FAISS and tuning a cosine-distance threshold to trade precision against recall; served event-grouped results to 300+ users via a FastAPI and React app.

Computer Vision Attention Monitoring System | *Python, OpenCV, MediaPipe, AWS (EC2, RDS, S3), Streamlit*

- Deployed a real-time attention monitoring system on AWS, processing 30 FPS webcam streams at <50ms/frame using 468-point facial landmarks and head-pose estimation across 10+ concurrent students.

LEADERSHIP & ACTIVITIES

IEEE Professional Communication Society, VIT Vellore, India

Vice Chairperson *March 2025 – March 2026*

- Directed chapter operations and recruitment, sustaining a 40% increase in active engagement; coordinated 3 cross-functional teams to run technical events on schedule and mentored 30+ junior members.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional, Oracle University August 2026